Evaluating controlled cooling therapy for the prevention of Chemotherapy Induced Peripheral Neuropathy (CIPN)

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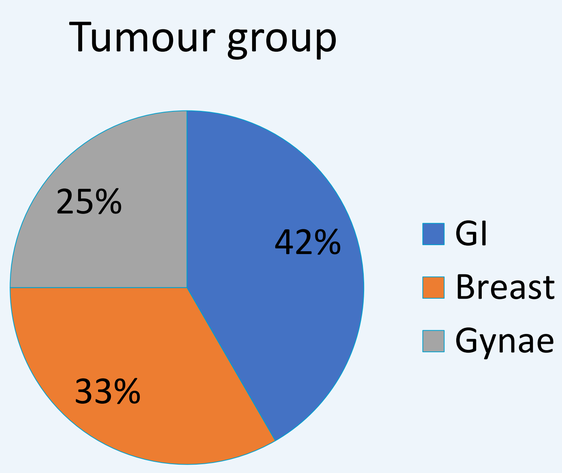
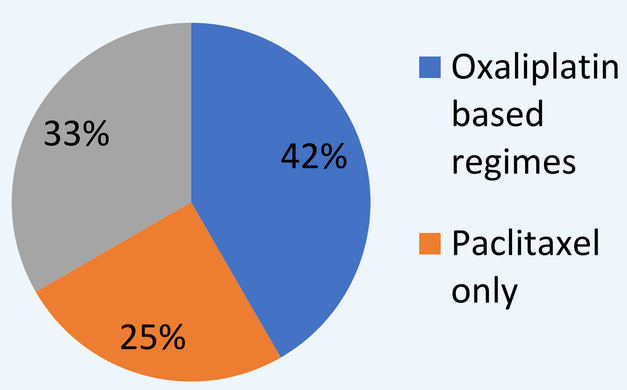
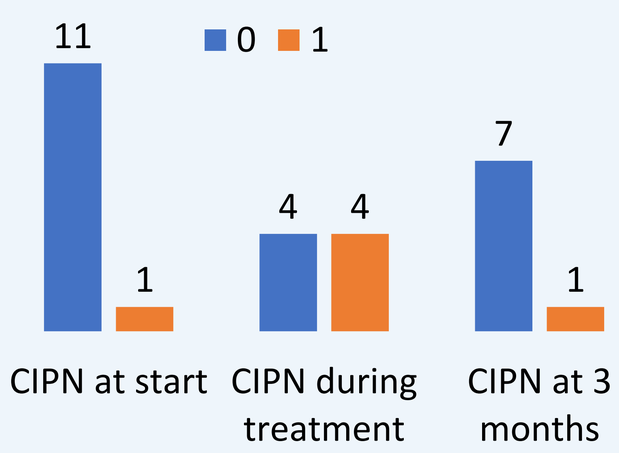
Introduction

- CIPN is a common side effect of chemotherapeutic drugs, particularly taxane and platinum-based drugs (Schaper et al, 2020).
- Symptoms range from acute and transient to chronic and painful, and they have a significant impact on the patient's quality of life. There can also be an impact on treatment schedules with dose reductions or cessation of treatment, which can impact long-term patient prognosis (Oneda et al, 2021).
- Growing evidence suggests that by using controlled thermotherapy (Hilotherm Chemocare), fewer patients develop grade 2 CIPN while finding the device comfortable to use (Coolbrandt et al, 2020; 2023).







Aims

- To evaluate the development of CIPN during chemotherapy treatment using common terminology criteria for adverse events (CTCAE).
- Secondary objectives are to assess the sustainability of the impact (at 3 months), identify alterations in the initial plan due to CIPN, and establish the level of adherence to the device.

Methods

- Observational prospective study. Data was collected over a 6-month period.
- Adult patients with breast, gynaecological, or gastrointestinal (GI) cancer who received carboplatin and paclitaxel in combination, paclitaxel alone, or oxaliplatin-based regimens were included.
- Patients treated with other neurotoxic chemotherapeutic agents, with CIPN, skin infection, or any of the stated contraindications of use were excluded.

Results

- 12 patients: 5 GI, 4 breast, 3 gynaecological cancer.
- SACT drugs: 5 oxaliplatin, 4 carboplatin/paclitaxel,3 paclitaxel.
- Of the 8 patients that adhered to the device: 4 did not develop CIPN and 4 developed grade 1 CIPN during treatment; after 3 months, 7 had no CIPN and 1 had grade 1 CIPN.
- Treatment modifications due to CIPN: 1
- 8 out of 12 adhered to the device: drop out mostly due to time wearing gloves.

Conclusion

- This service evaluation provided valuable data regarding the development of CIPN using a cooling device.
- Controlled cooling appears feasible and safe. Most patients developed only mild CIPN, contrasting with experiential data in non-cooled patients.
- Adherence rates suggest good acceptability, particularly in shorter treatments.
- Findings support further investigation.

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References

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